Higman–Sims groups from intrinsic collections of small types

Private finite application

Let $G = \mathrm{HS}$ in the acquired faithful action of degree 100. The complete coset catalogue and the identified automorphism group $N = \mathrm{Aut}(G)$, of order 88 704 000, are retained inputs. We use the source coordinates of the archived supplement to D. Leemans, *The residually weakly primitive geometries of HS*, Australas. J. Combin. **33** (2005), 231–236. The claims below concern the explicitly listed families only.

Restriction and faithfulness

Select all types having cardinality in a specified finite set. The collection is intrinsic, including all repeated cardinalities. Its induced incidence graph has one additional marker per type. An object is coloured by its type cardinality; marker colours are distinct from object colours and otherwise also depend only on cardinality. A marker is joined to exactly the objects of its type. Thus the colour-preserving graph group is exactly the correlation group of the selected incidence system, with no artificial distinction between types of the same size.

Lemma 1. *Suppose the image E of G on the selected objects is faithful and is characteristic in their full induced graph group. Suppose every typed automorphism fixing the selected objects fixes all objects of the original geometry. Then every correlation of the original geometry normalizes its specified G action.*

Proof. A correlation c restricts to the intrinsic selected system. For every $g \in G$, characteristicity gives $h \in G$ whose restriction equals that of cgc^{-1}. The typed automorphism $h^{-1}cgc^{-1}$ fixes every selected object, so is trivial by hypothesis. Consequently $cgc^{-1} = h \in G$. □

Here faithfulness is checked through the actual coset action and successive incidence descriptions. Write P_i for the base parabolics and b_j for the selected base objects. The first description of a type-i object is its tuple of shadows on the selected types. At the base object this is $(b_j^{P_i})_j$. Its stabilizer in G is computed as an intersection of actual set stabilizers. It equals P_i exactly when this description is injective within type i.

If some types are not yet distinguished, use a type k whose description has already been proved injective. Append to the description of type i the set of descriptions of its type-k neighbours. The base set is the orbit of the base description of type k under P_i; its size is checked to equal $[P_i : P_i \cap P_k]$. Intersecting the old stabilizer with this new set stabilizer gives the refined stabilizer. Each step uses only a previously established type. Induction proves that a typed automorphism fixing the selected objects fixes every described object. When all final stabilizers are P_i, the hypothesis of the lemma follows. This is the already acquired iterative-shadow argument, applied to joint selected actions.

Literal finite identifications

For each selected type the saved ordered generator images define an actual homomorphism from the acquired G to the joint permutation group E. The verifier checks that it is bijective. The base vertex has stabilizer equal to the image of P_i, and its orbit is the entire type. Hence $P_i g \mapsto b_i^g$ is the required coset bijection; no agreement of historical vertex labels is assumed.

For each ordered pair of selected types, the neighbours of the base vertex are checked to be exactly the appropriate parabolic orbit. Every generator preserves every coloured adjacency row. Equivariance then gives all incidences and nonincidences. The complete nauty group is a computational premise; every returned generator is also checked on every coloured row. GAP verifies the exact order of the generated group D, containment $E \leq D$, and $D' = E$. This last equality establishes the characteristic subgroup needed in the lemma, rather than inferring it from a numerical order alone.

The shadow verifier reconstructs all initial orbits and stabilizers, then every recorded refinement in its actual chronological order. It checks the exact finite descriptions, the complete neighbour orbit, the saved stabilizer generators, and equality to every parabolic. In all nine applications below, the initial shadows already distinguish every type: there are no refinement steps. Successful candidate output or agreement with a source order is not substituted for these checks.

From normalization to full groups

Let $B = \bigcap_i P_i$, and let S, T be respectively the ordered and setwise family stabilizers in the complete $N = \mathrm{Aut}(G)$. If $P_i^t = P_{\pi(i)}$, the rule $P_i g \mapsto P_{\pi(i)} g^t$ is an incidence isomorphism. Conversely a correlation normalizing G can be moved by G to fix the base chamber and then has precisely this form. Therefore

$$|\mathrm{Aut}(\Gamma)| = |G : B||S|, \qquad |\mathrm{Cor}(\Gamma)| = |G : B||T|.$$

All type permutations are tested. Orders and intersection orders reject only impossible permutations; successive complete subgroup transporters handle the rest, and every positive lift is checked on the whole family. This is the acquired normalizer argument, with literal new families.

The computation also checks $S \cap G = B$ and the actual quotient $(T \cap G)/B$. The latter is the centralizer of the specified G in the full correlation group. When it is trivial, the full group identifies with its conjugation image in N. When it has order two, it is central. If the typed group is G and the full group has twice its order, the central factor has trivial intersection with G and their product has full order. This proves $G \times C_2$, not merely an order equality.

Theorem 2. *For the nine acquired families in the table, the full type-preserving automorphism group is G. Their full correlation groups are as shown. Here v counts selected objects and type markers, $d = |D|/|G|$, and $t = |T|/|B|$; the source label uses the acquired exact coordinate map.*

Family	Source	v	d	$\lvert B\rvert = \lvert S\rvert$	t	$\mathrm{Cor}(\Gamma)$
24	3.12	1454	1	72	1	G
25	3.15	2555	2	72	2	$G \times C_2$
38	4.28	1454	1	4	1	G
57	4.19	706	4	8	2	$G \times C_2$
58	4.17	1454	1	8	1	G
59	4.16	1807	2	8	2	$G \times C_2$
63	4.22	1807	2	8	1	G
64	4.21	1454	1	8	1	G
65	4.20	1454	1	8	1	G

Proof. For each row the literal checks establish $D' = E \cong G$, injectivity of every initial shadow, and the complete ordered and setwise family stabilizers. The preceding restriction lemma and normalizer formulas therefore give the full groups. When $t = 2$, the verified centralizer $(T \cap G)/B$ has order two, so the direct-product argument applies. The larger selected graph group for family 57 is only an upper-bound vehicle: the full geometry has $t = 2$, not $d = 4$. □

Finite scope. Families 21, 22, 37, 39, 54 and 56 are explicitly left open by this component; their individual 60-second model attempts were interrupted. Candidate attempts that reached their individual deadlines remain unproved cases. In those cases the recorded normalizer groups are only lower bounds. No ambient subgroup or geometry enumeration is repeated; source names, numerical classification and software completeness remain explicit premises. No whole-source completion, new conjecture, formalization or measurement claim is made.